Title: Bias-Audited LLM Resume Screening with Explainable, Anonymization-Aware Ranking

Project Group Members

Guide: P. Srilakshmi

S.No.	Roll No.	Student Name
1	2420030089	B.Sai Madhav Rao
2	2420030078	Jitheswar
3	2420090072	Prashanth

Abstract

Automated resume screening promises to save recruiter time, but two problems undercut its trustworthiness: rankings are rarely explainable, and language models are known to pick up on identity signals - name, gender markers, university prestige - that have nothing to do with fitness for a role. This project builds an **LLM-based Resume Screening system that ranks a pool of resumes against a single job Role and pairs every ranking with a measured, reproducible bias audit**, rather than a bias disclaimer bolted on after the fact.

Given a Role (title plus free-text description) and an Applicant Pool, the system derives a fixed four-criterion Rubric (skills, experience, education, domain fit) once per Role, scores every resume against that frozen rubric through an LLM client, and produces a ranked Candidate Recommendation in which every candidate is present and every position carries a Justification naming what the resume evidences and what it lacks. No candidate is ever filtered out - screening only orders. Two retrieval conditions are evaluated: a standard 300-resume pool at a fixed 20% relevance ratio, and a full-corpus condition (~2,400 resumes) where retrieval is load-bearing rather than decorative.

Fairness is measured, not assumed. Identity Signals (name, gender markers, graduation year) are stripped by an Anonymization stage before scoring, while a Contested Signal - university - is deliberately left in the resume so its effect can be measured rather than engineered away. A Bias Audit generates counterfactual resume pairs that differ in exactly one signal at a time, along three axes (name/gender, name/ethnicity, university), and reports the resulting change in rank position and score, with anonymization on and off, so its mitigation effect is a measured delta rather than a claim. Ranking quality is evaluated against category-based relevance labels and against a small hand-ranked slice judged by a human, so the coarse automatic label and human judgment can be compared directly. The result is a screening pipeline whose rankings are explainable by construction and whose fairness claims are backed by a repeatable audit rather than a single anecdote.

Datasets

Dataset	Source
Kaggle Resume Dataset (livecareer)	Resume text (Resume_str) and job category (Category) columns, used as the applicant pool corpus (~2,400 resumes)
Synthetic counterfactual variants	Generated in-pipeline by injecting one identity signal (name/gender, name/ethnicity, university) at a time into a base resume
Hand-ranked slice	A small subset of the pool ranked by a human judge, used to validate the automatic relevance label

Approach

```
                         MULTIPLE INPUTS
              +-------------------+-------------------+
              v                   v                   v
          Job Role          Applicant Pool       Full-Corpus
        (title + desc)        (300 @ 20%)          (~2,400)
              |                   |                   |
              +-------------------+-------------------+
                                            v
                        Rubric Derivation
                        (frozen per Role)
                                            v
                           Anonymization
                 (Identity Signals stripped;
                  Contested Signal retained)
                                            v
                            LLM Scoring
              (skills / experience / education
                           / domain fit)
                                            v
                  Ranked Candidate Recommendation
              (every candidate + Justification)
                      +--------------+--------------+
                      v                             v
               Bias Audit                    Evaluation
        (counterfactual pairs,        (category relevance,
         3 axes, rank/score             hand-ranked slice
          deltas, anon on/off)               agreement)
                      +--------------+--------------+
                                            v
                  Figures / Analysis Notebook
```